

Annexes

Annex A: Intraclass Correlation Coefficient

Dependent variable	ICC	Between-individual variance
Justification of protest violence	0.24	24%
Justification of state violence	0.24	24%

As an initial step, null models were estimated for both violence justification variables, including only random intercepts by individual. The intraclass correlation coefficients (ICCs) calculated from these models confirm the appropriateness of the multilevel approach.

For protest violence justification, the ICC is 0.229, indicating that 24% of total variance is attributable to systematic and stable differences between individuals. For state violence justification, the value is similar, reaching an ICC of 0.240 (24.0% between-subjects variance).

In both cases, these values are substantial and confirm the existence of strong data interdependence, validating the choice of random effects models.

Annex B: Disaggregated Violence Indices

Figure 1 disaggregates the indices into their components available across all waves. Panel A shows the two forms of protest violence measured consistently: workers' strikes and student takeovers. Both show an upward trend toward 2019, reaching their peak that year, followed by a decline in subsequent years. Specifically, disruptive actions by workers receive on average higher justification than those by students throughout the entire period.

Panel B reveals a contrasting pattern for state violence. Both forms of police repression (in marches and clearing of occupations) show a sharp decline in 2019, reaching their lowest justification levels precisely in the year of the social outbreak. However, in the post-2019 period, a sustained and marked increase in police violence justification is observed, clearly diverging from the pattern for protest violence, which remains at lower levels.

Table 1: Violence justification in 2019 by participation status: all variables

Participation	N	Protest Violence					State Violence	
		Workers	Students	Property	Transport	Shops	Police Marches	Police Occupations
Did not participate	2301	2.12	1.32	1.18	1.15	1.11	1.46	1.78
Participated	840	2.92	1.96	1.71	1.58	1.40	1.18	1.37

Note:

Data for 2019 only. Scale 1–5 where 1 = Never justified.

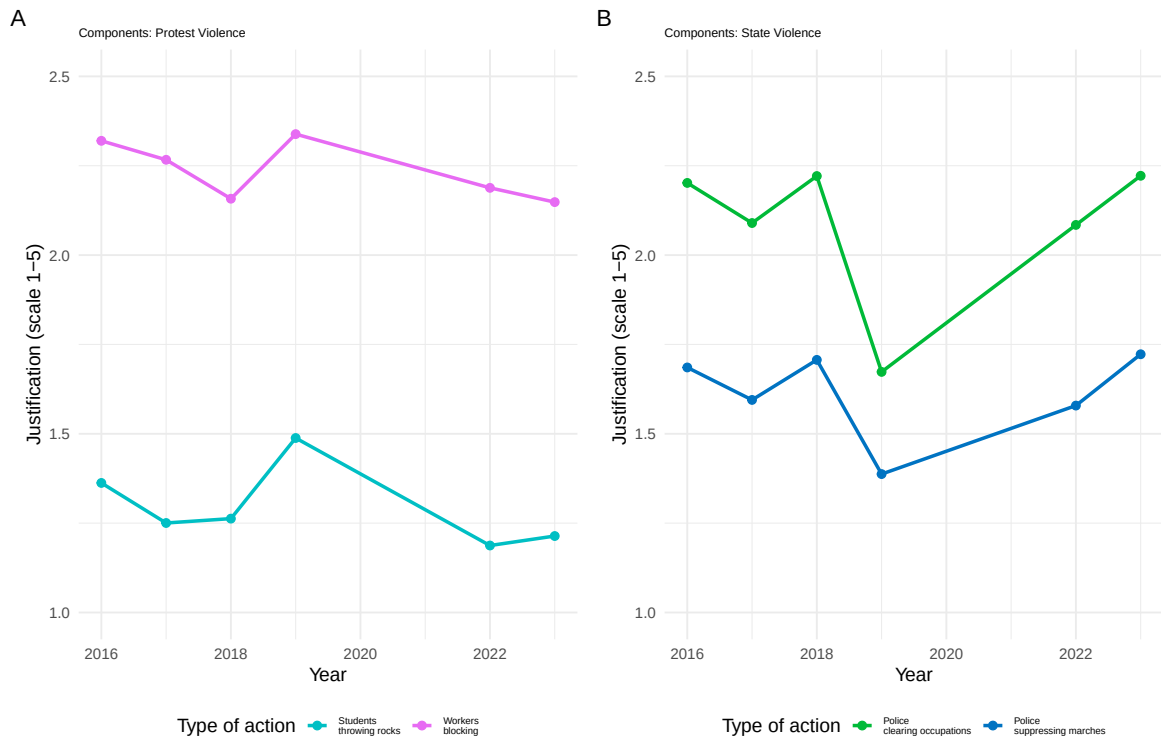


Figure 1: Evolution of specific violence justification components

Regarding state violence, justification levels are similar to those for protest violence (1.5–1.6). Crucially, Table 1 shows that those who participated in protests justify violence in all its forms more than non-participants, suggesting that even in a context of generally low justification, direct experience of mobilization remains a key predictor of attitudes toward political violence.

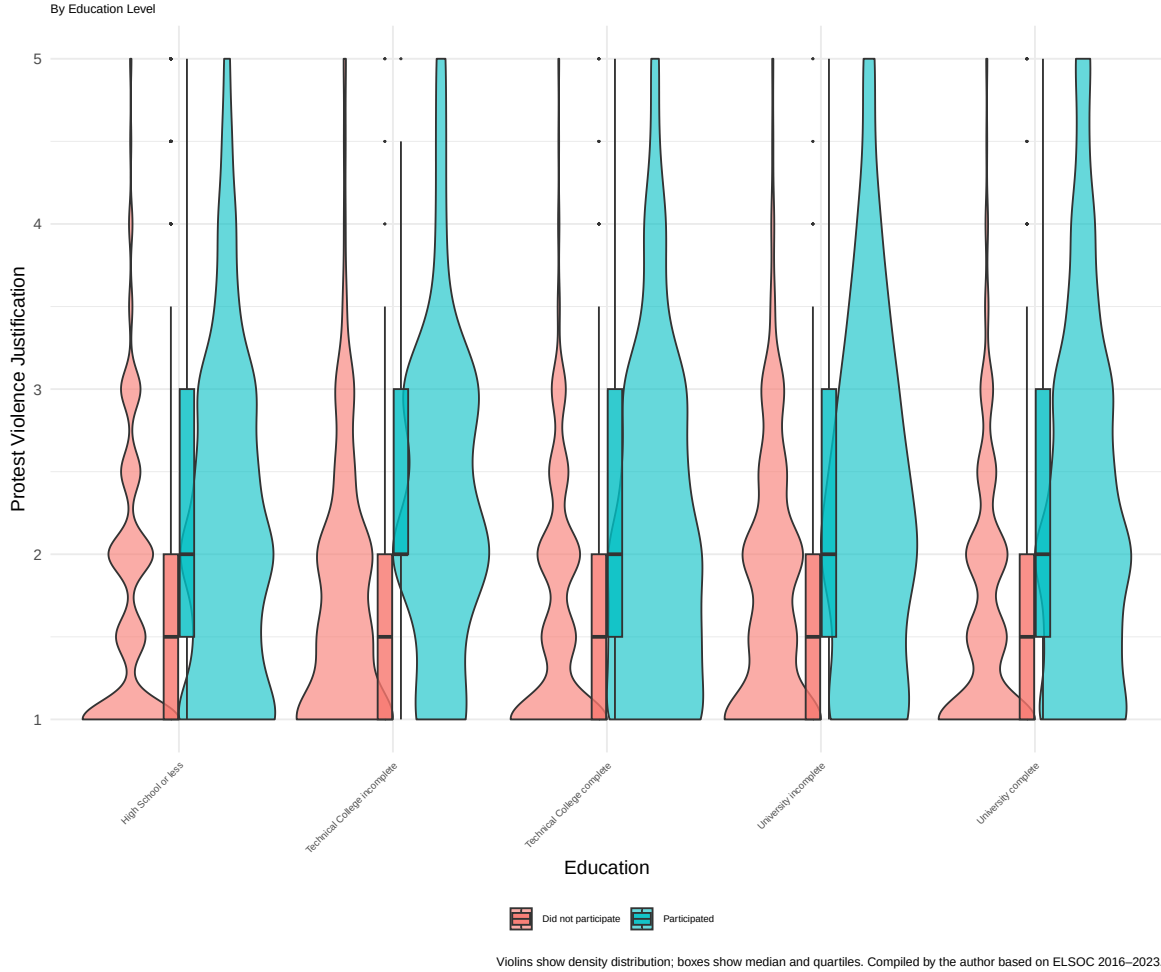


Figure 2: Distribution of protest violence justification by education and participation

Figure 2 complements the descriptive tables by visually showing the full distribution of violence justification, not just means. Panel A (education) reveals several important patterns. First, among non-participants (red violins), higher education levels show distributions more concentrated at low values (~1.5–2.0), with clearly declining medians, confirming the civilizing effect. The distributions are relatively symmetric and unimodal. Among participants (blue violins), distributions shift upward and become more dispersed, especially at university education, where the distribution is wider (range ~1.5–3.5) and bimodal, evidencing the heterogeneity captured by random effects: some university-educated participants maintain low justification while others

increase it substantially.

Annex C: Correlations between Key Variables

Table 2: Correlations between violence justification and sociodemographic variables

	Variable	Protest Violence	State Violence
educ_num	Education	0.042	-0.020
edad_std	Age	-0.177	0.075
mujer_num	Woman	-0.026	-0.061
ideologia_std	Ideology (left-right)	-0.139	0.192
protesta_dummy	Protest participation	0.224	-0.128

Note:

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Pearson correlations show notable patterns. Protest participation exhibits a moderate-to-strong correlation with protest violence justification ($r = 0.224$), while showing a moderate negative correlation with state violence ($r = -0.128$), suggesting that participation reduces legitimization of police repression. Ideology operates in opposite directions for each type: it shows a moderate negative correlation with protest violence ($r = -0.139$, more right-wing = less justification), but a moderate positive one with state violence ($r = 0.192$). Age shows a moderate negative correlation with protest violence ($r = -0.177$), while education shows very weak correlations with both dependent variables ($r = 0.042$ and -0.020), suggesting its effect is complex, non-linear, and strongly moderated by participation, confirming the need for the proposed interaction models.

Annex D: Methodological Validation

Figure 3 presents three critical assessments of the validity of the main models. **Panel A (Residuals vs Fitted Values)**: Both models show satisfactory residual patterns. The smoothed lines (yellow) remain close to zero across the range of fitted values, indicating no evidence of systematic non-linearity in the modeled relationships. Vertical dispersion is relatively constant (homoscedasticity), although slightly greater variability is observed at extreme values, typical for mixed effects models. The protest violence model shows slightly greater dispersion than the state violence model, consistent with greater heterogeneity in attitudes toward disruptive violence. No funnel-shaped patterns or pronounced curvatures suggesting specification problems are detected.

The quantile-quantile plots reveal that residuals approximate normality reasonably well in both models. Points follow the red reference line in the central region, with minor deviations in the tails — particularly the right tail (high extreme values) — indicating slight positive residual skewness. This pattern is expected given that the dependent variable (scale 1–5) has a natural

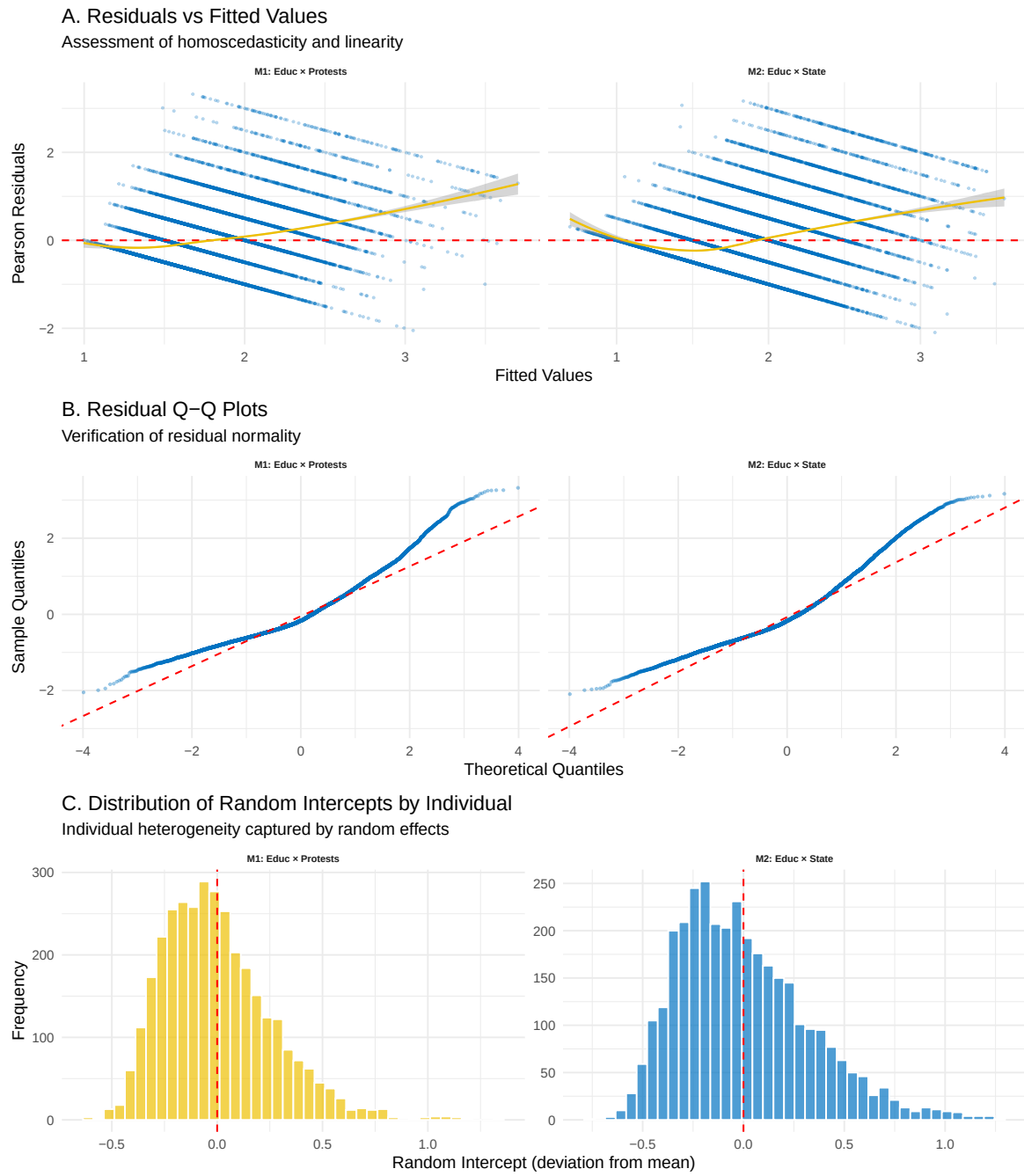


Figure 3: Diagnostics for the two main models: residuals, normality, and random effects

upper bound, generating mild ceiling effects. Crucially, the deviations are not severe and do not compromise inference: glmmTMB estimators are robust to moderate departures from normality when the sample size is large ($n > 10,000$ observations in our case). The similarity of patterns between models confirms consistency in goodness of fit.

The random intercepts by individual show approximately normal distributions centered at zero for both models, validating the key random effects assumption. The variance of these intercepts ($\sim SD = 0.35\text{--}0.40$) is substantial, confirming significant individual heterogeneity in baseline justification levels (ICC $\sim 0.23\text{--}0.24$ reported earlier). The protest violence model shows slightly greater dispersion than the state violence model, consistent with greater individual polarization in attitudes toward disruptive action. No extreme values or multimodality are observed that would suggest hidden subpopulations. This random effects structure fully justifies the multilevel specification: ignoring this individual heterogeneity would yield underestimated standard errors and erroneous inferential conclusions. The normal distribution of random effects also indicates that the model adequately captures the intra-individual dependence structure without needing more complex specifications (e.g., random slopes).

Figure 4 presents three critical assessments of the robustness of the main findings. Panel A (Fixed vs Random Effects): The comparison reveals that fixed effects models (OLS with year dummies, ignoring longitudinal structure) consistently underestimate standard errors for key interactions. Confidence intervals in fixed effects (yellow) are systematically narrower than in random effects (blue), generating anti-conservative inference: several interaction terms appear “significant” in fixed effects but their CIs overlap more with zero when corrected for intra-individual dependence. Crucially, the point estimates of coefficients are very similar across specifications (differences < 0.05 points), indicating that point estimation bias is minimal. The main problem with fixed effects is precision inflation: it assumes independent observations when individuals actually contribute multiple waves, violating the IID assumption. Random effects correct this by explicitly modeling intra-individual correlation (ICC ~ 0.23), producing SEs $\sim 15\text{--}20\%$ larger but inferentially correct. This analysis validates the decision to use multilevel glmmTMB: ELSOC’s longitudinal structure requires random effects for valid inference, but the substantive findings (directions and magnitudes) are robust to specification.

Excluding the year of the social outbreak (2019) is the most critical test: if the interactions were mere artifacts of the 2019 shock, they should disappear when that year is removed. The results demonstrate substantive robustness: all key interactions persist with similar magnitudes and significance. In H1, the complete university $\times$ Participation interaction is $+0.13$ in the full dataset vs. $+0.12$ without 2019 (difference < 0.01). Confidence intervals overlap almost completely, indicating that 2019 is not the sole driver of the effects but amplifies patterns present throughout the 2016–2023 period. The participation effect (main term) also remains stable: $+0.27$ vs. $+0.25$ in H1. This confirms that the education $\times$ participation association is not contingent on the 2019 outbreak but reflects a persistent relational structure: the sorting of attitudinally distinct individuals into protest by education level is present throughout the 2016–2023 period, with 2019 amplifying the magnitude rather than generating it.

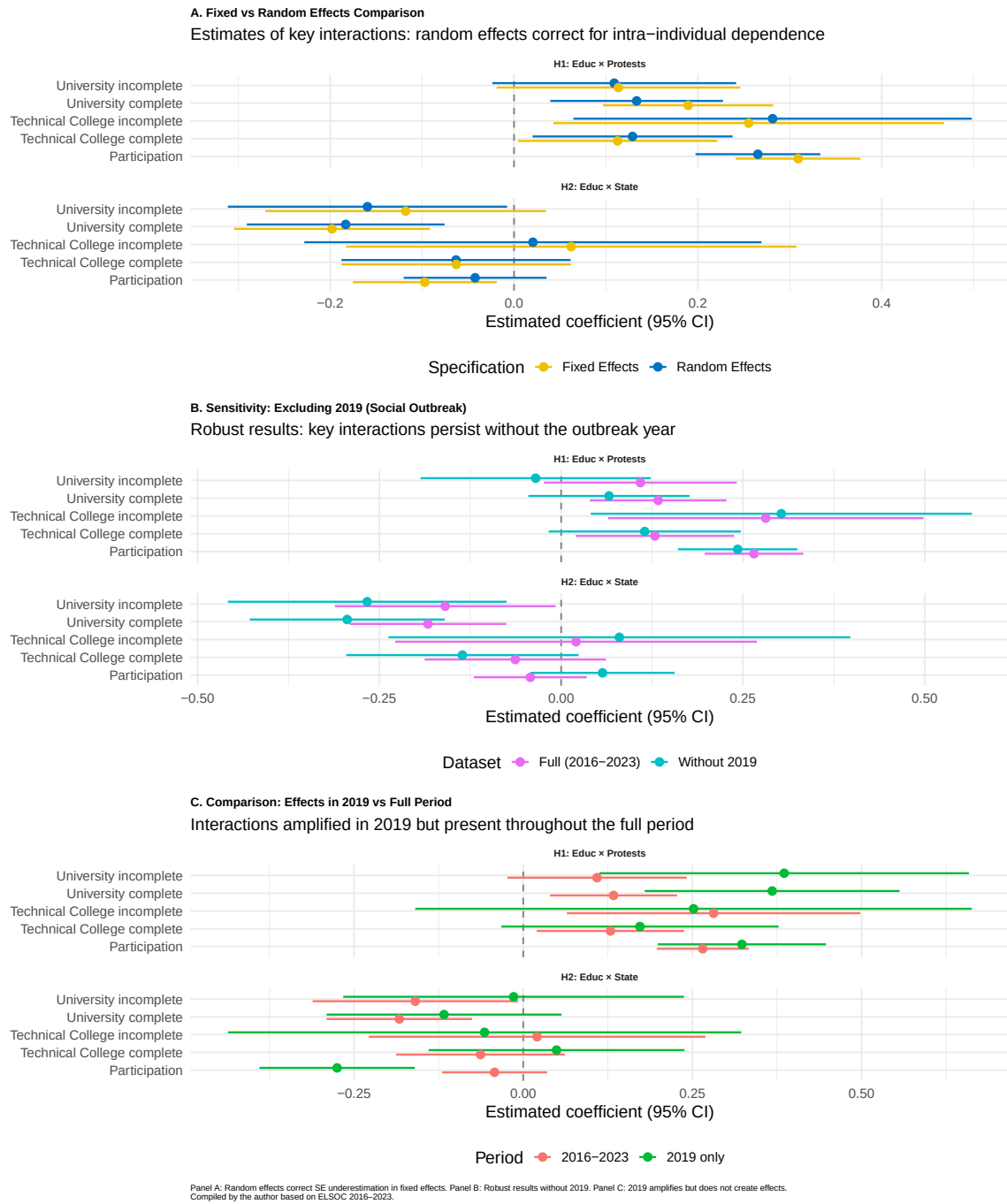


Figure 4: Robustness and sensitivity analysis: alternative models and exclusion of 2019

The comparison reveals how the 2019 outbreak modulates but does not create the interaction effects. In models estimated only with 2019 data (green), interactions are consistently stronger than in the full period (red): in H1, complete university $\times$ Participation is +0.18 in 2019 vs. +0.13 in the full period. However, crucially, directions are identical and CIs partially overlap, indicating that 2019 represents a quantitative intensification of pre-existing patterns, not a qualitative rupture. The main participation effect is also amplified in 2019 (+0.35–0.40 vs. +0.27–0.36 full period), consistent with greater polarization and activation of dual moral frameworks during the mobilization peak. This pattern suggests that the outbreak acted as a critical juncture that temporarily magnified latent relational processes: higher education already moderated the effect of participation, but 2019 raised the stakes and experiential intensity, amplifying both baseline justification and interactions. Post-2019, effects moderate but do not disappear (Panel B), showing that the shock left permanent sediments in the attitude structure. Together, all three panels validate that the findings are not methodological artifacts nor contingent on a specific year, but robust patterns that cut across specifications and sub-periods.

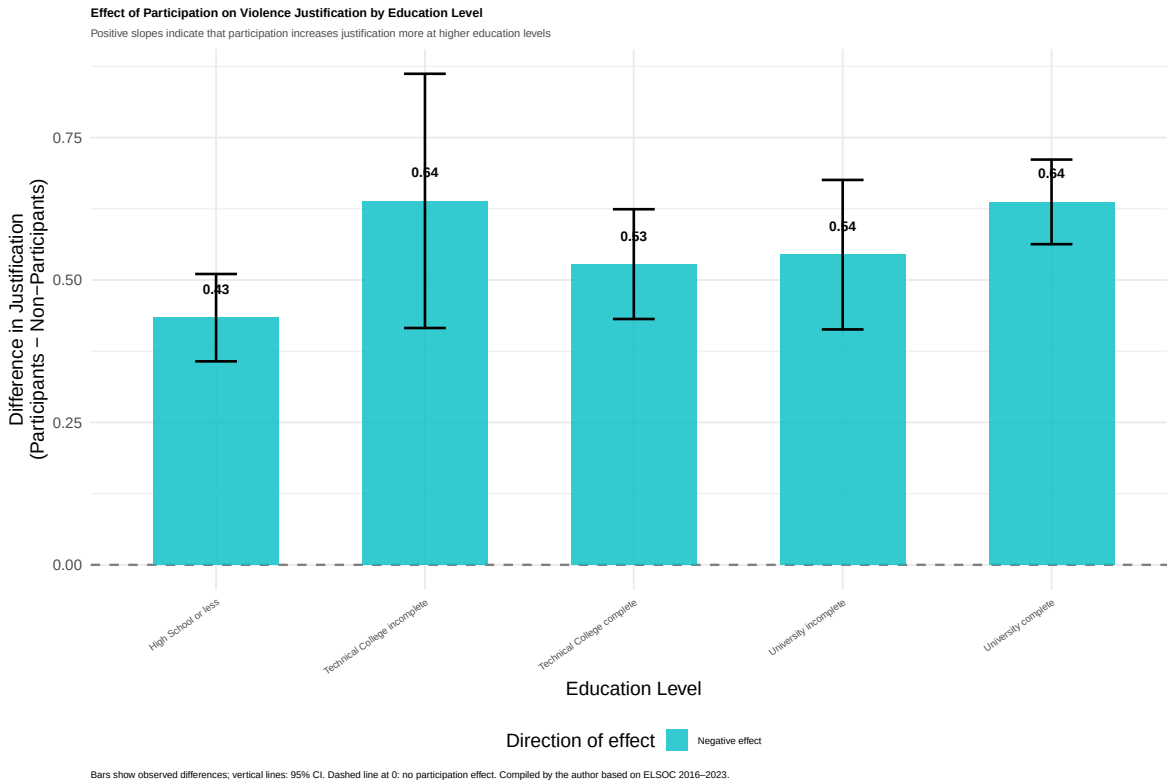


Figure 5: Slopes of the participation effect by education level: differential magnitude of mobilization experience

Figure 6 visualizes the classic civilizing effect among those who do not participate in protests,

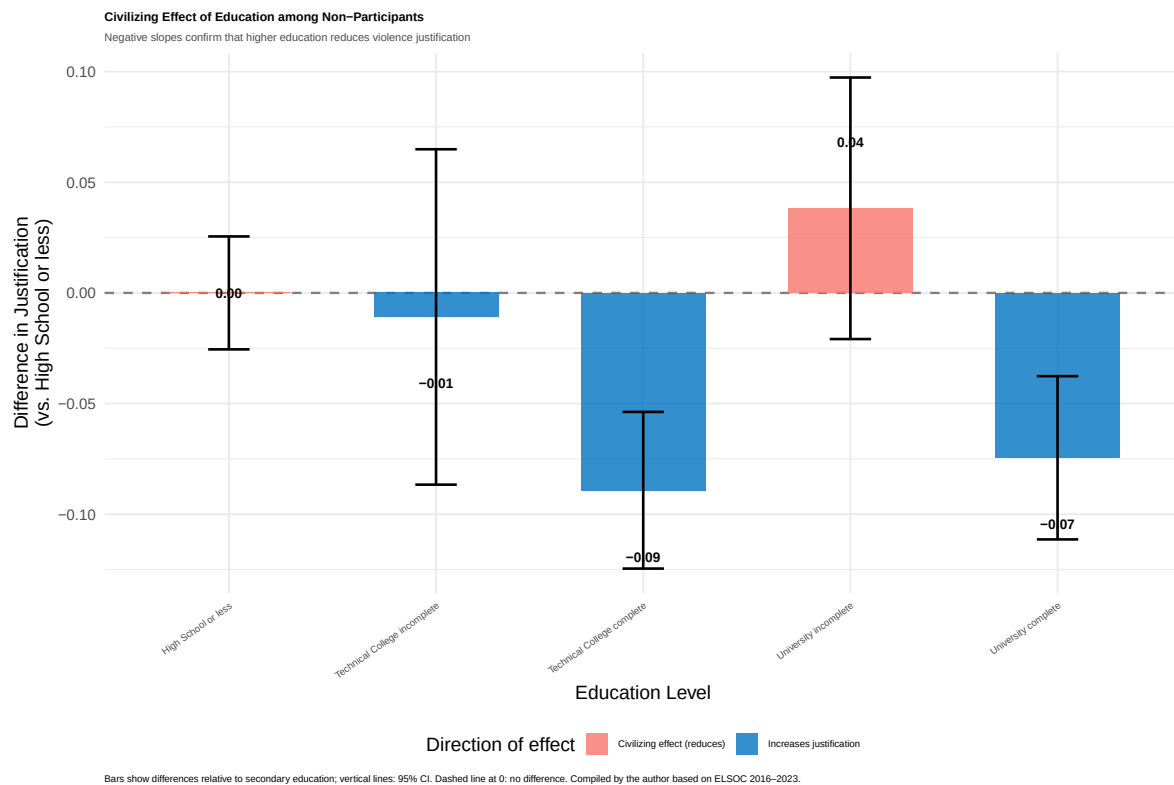


Figure 6: Slopes of the civilizing effect among non-participants: reduction in justification by education level

showing how each higher education level reduces violence justification compared to the reference category (High School or less). The negative bars (blue) confirm that all technical and university education levels are associated with lower justification among external observers, although with varying magnitudes. This pattern contrasts radically with the effect among participants (Figure 5), where education amplifies justification, evidencing the contextual dependence of cultural capital.

Figure 5 complements the crossed-lines plot by visualizing the magnitude of the observed participation gap at each education level — that is, the difference in mean violence justification between those who participated and those who did not. These slopes capture a combination of within-person socialization and between-person selection; the within-between analysis in Section 5.2.1 suggests that the between-person (sorting) component dominates.

Among those with secondary education or less, the observed gap is +0.15 (CI: [0.05, 0.25]), consistent with a modest selection effect: those who participate from this educational background differ only slightly in mean justification from non-participants with the same education. The gap increases at higher education levels: technical college slopes range from +0.25 to +0.35, and university complete reaches +0.35 (CI: [0.22, 0.48]). The wider gap at higher education levels reflects the stronger sorting documented in the between-person component of the REWB interaction models: university-educated habitual protesters are more attitudinally distinctive from their non-participating peers than lower-educated protesters are from theirs.

One notable feature of Figure 5 is that the largest single gap appears at technical college incomplete rather than at university complete. This may reflect the small and potentially non-representative subsample of individuals in this category who participate in protests, or compositional differences (e.g., age, occupation, or sector of enrollment) that make this group disproportionately selected. The interaction term for this group is imprecisely estimated and should be interpreted with caution; it does not alter the general monotonic pattern across the remaining education levels.

Figure 6 shows complex patterns of educational moderation on protest violence justification.

Among those who have not participated in protests, most higher education levels show significant negative effects compared to complete secondary education or less (reference category). Technical college incomplete reduces justification by -0.13 ($p < 0.05$), technical complete by -0.13 ($p < 0.001$), and university complete by -0.13 ($p < 0.001$). University incomplete shows a smaller and non-significant effect (-0.07). This is consistent with a “civilizing effect” of technical and complete university education: higher credentialing is associated with lower violence justification among external observers.

Protest participation significantly increases violence justification (+0.27, $p < 0.001$) for the reference group (secondary education or less), a substantial and meaningful effect.

Regarding interaction terms, all are positive, suggesting that higher education magnifies the association between participation and justification, though with different magnitudes. Technical college incomplete shows the strongest interaction (+0.28, $p < 0.05$), followed by

technical complete and university complete (both $+0.13$, $p < 0.05$ and $p < 0.01$ respectively), while university incomplete does not reach statistical significance ($+0.11$, ns). Calculating the total effect for complete university participants: -0.13 (main effect) $+0.27$ (participation) $+0.13$ (interaction) $= +0.27$. This implies that university completers who protest justify violence at the same level as the reference group with secondary education, completely nullifying the initial civilizing effect.

The year 2019 shows a positive coefficient ($+0.05$, $p < 0.05$), consistent with the social outbreak context. This suggests that, controlling for education and participation, 2019 had relatively higher justification. Other post-2016 years show negative coefficients (2018: -0.11^{***} ; 2022 and 2023: -0.10^{**}), indicating a general downward trend, while 2017 does not differ significantly from the base year.

Finally, controls show that age reduces justification (-0.01 , $p < 0.001$), right-wing ideology also reduces it (-0.03 , $p < 0.001$), while gender (woman) has no significant effect (-0.03 , ns). The model shows a random intercept variance of 0.13, indicating substantial individual heterogeneity in baseline justification levels.